

# CM0845 Logic Other Topics

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# Some Related Subjects

- Logic monism or logic pluralism (Logic or logics?)
- Time matters (algorithmic complexity)
- When you only have a hammer, everything looks like a nail (paradigms of programming)
- Help from the machines (theorem provers)

# Logic or Logics?

See slides in the course web page.

# Algorithmic Complexity

## Comparison of several time complexity functions

| $f(n)$   | 10       | 50                     | 100                    |
|----------|----------|------------------------|------------------------|
| $\log n$ | 2.3 sec  | 3.9 sec                | 4.6 sec                |
| $n$      | 10 sec   | 50 sec                 | 1.7 min                |
| $n^2$    | 1.7 min  | 41.7 min               | 2.8 h                  |
| $2^n$    | 17.1 min | 358.001 c              | $4 \times 10^{20}$ c   |
| $3^n$    | 16.4 h   | $2.3 \times 10^{14}$ c | $1.6 \times 10^{38}$ c |
| $n!$     | 42 d     | $9.7 \times 10^{54}$ c | $3 \times 10^{148}$ c  |

# Algorithmic Complexity

## Example (3-SAT: An intractable problem)

To determine the satisfiability of a propositional formula in conjunctive normal form where each disjunction of literals is limited to at most three literals.

The problem was proposed in Karp's 21 NP-complete problems.<sup>†</sup>

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<sup>†</sup>Karp (1972). Reducibility Among Combinatorial Problems.

# Algorithmic Complexity<sup>†</sup>

## 3-SAT better deterministic algorithmic complexity

| Year | Complexity   | Reference                                                                                |
|------|--------------|------------------------------------------------------------------------------------------|
| 2010 | $O(1.439^n)$ | Kutzkov and Scheder (2010). Using CSP to Improve Deterministic 3-SAT (unpublish).        |
| 2008 | $O(1.465^n)$ | Scheder (2008). Guided Search and a Faster Deterministic Algorithm for 3-SAT.            |
| 2004 | $O(1.473^n)$ | Brueggemann and Kern (2004). An Improved Deterministic Local Search Algorithm for 3-SAT. |

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<sup>†</sup>Moser and Scheder (2011). A Full Derandomization of Schönig's k-SAT Algorithm.

# Algorithmic Complexity

## 3-SAT simulation

Machines from: [www.top500.org](http://www.top500.org)

Petaflops (PFs):  $10^{15}$  floating-point operations per second

| Date            | Machine     | PFs   | 100     | 150    | 200                 |
|-----------------|-------------|-------|---------|--------|---------------------|
| 2013-06 2015-10 | Tianhe-2    | 33.86 | 2.0 sec | 16.0 y | $4.1 \times 10^7$ c |
| 2012-06         | Blue Gene/Q | 16.32 | 4.1 sec | 33.1 y | $8.5 \times 10^7$ c |
| 2011-06         | K computer  | 8.16  | 8.1 sec | 66.2 y | $1.7 \times 10^8$ c |

# Some Paradigms of Programming

**Imperative:** Describe computation in terms of state-transforming operations such as assignment. Programming is done with statements.

**Logic:** Predicate calculus as a programming language. Programming is done with sentences.

**Functional:** Describe computation in terms of (mathematical) functions. Programming is done with expressions.

## Examples

Imperative  $\left\{ \begin{array}{l} \text{C} \\ \text{C}++ \\ \text{Java} \end{array} \right.$

Logic  $\left\{ \begin{array}{l} \text{CLP(R)} \\ \text{Prolog} \end{array} \right.$

Functional  $\left\{ \begin{array}{l} \text{ML} \\ \text{Erlang} \\ \text{Pure} \left\{ \begin{array}{l} \text{Clean} \\ \text{Haskell} \\ \text{Idris} \end{array} \right. \end{array} \right.$



# Theorem Provers

See slides in the course web page.

# References

- Tobias Brueggemann and Walter Kern (2004). An Improved Deterministic Local Search Algorithm for 3-SAT. Theoretical Computer Science 329.1–3, pp. 303–313. DOI: [10.1016/j.tcs.2004.08.002](https://doi.org/10.1016/j.tcs.2004.08.002) (cit. on p. 6).
- Richard M. Karp (1972). Reducibility Among Combinatorial Problems. In: Complexity of Computer Computations. Ed. by Raymond E. Miller and James W. Thatcher. Plenum Press, pp. 85–103. DOI: [10.1007/978-1-4684-2001-2\\_9](https://doi.org/10.1007/978-1-4684-2001-2_9) (cit. on p. 5).
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- Robin A. Moser and Dominik Scheder (2011). A Full Derandomization of Schönning's k-SAT Algorithm. In: Proceedings of the Forty-third Annual ACM Symposium on Theory of Computing (STOC 2011), pp. 245–252. DOI: [10.1145/1993636.1993670](https://doi.org/10.1145/1993636.1993670) (cit. on p. 6).

# References

Dominik Scheder (2008). Guided Search and a Faster Deterministic Algorithm for 3-SAT. In: Proc. of the 8th Latin American Symposium on Theoretical Informatic (LATIN 2008). Ed. by Eduardo Sany Laber et al. Vol. 4957. Lecture Notes in Computer Science. Springer, pp. 60–71. DOI: [10.1007/978-3-540-78773-0\\_6](https://doi.org/10.1007/978-3-540-78773-0_6) (cit. on p. 6).